



An Investigative Study on Recent Sharpness- and Flatness-Based Optimizers for Enhanced Self-Supervised Speaker Verification

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Abstract

Flat minima have recently proven effective for improving generalization and robustness against label noise. While Adam and Stochastic gradient descent (SGD) remain the most popular optimizers in deep learning, other optimizers such as Sharpness-Aware Minimization (SAM), simultaneously minimizing loss value and minimizing loss sharpness and/or improving flatness, have proven to be more effective across a variety of tasks. Motivated by this, we present the first comprehensive study of general-purpose optimizers (ADOPT, AdEMAMix) and sharpness-aware variants (SAM, ASAM, GAM, GSAM) for the supervised and self-supervised speaker verification tasks. We also explore different regularization methods such as weight decay, EMA, and SWITCH EMA to enhance robustness and learn flat optima without extra optimization costs. Our experimental results highlight the significant impact of the optimizer choice on generalization, achieving state-of-the-art performance in self-supervised speaker verification.

Index Terms: Optimizers, label noise robustness, sharpness-aware minimization, self-supervised speaker verification

1. Introduction

Automatic speaker verification (ASV) as one of the most convenient means of biometric recognition [1], uses the voiceprint of a speaker to verify his or her identity. Based on known utterances of a speaker, the speaker verification (SV) task aims to identify whether a speaker is a legitimate user or an imposter. Although advances in self-supervised learning (SSL) techniques have helped to improve self-supervised SV leveraging large unlabeled datasets [2–5], label noise can still dramatically impact the final accuracy of predictions and generalization due to incorrect representations. As large-scale datasets are typically collected from the internet within automatic pipelines [6, 7], these memorization effects [8] can also impact supervised training. Therefore, designing robust and effective mechanisms to select reliable labels [9] for training and/or to curate these noisy annotations is a crucial task [10].

Many cleansing methods exist in the label noise and data pruning literature [11–14]. They rely in general on one or multiple stages of label noise cleansing followed by a supervised training stage that uses the curated training set. Besides, iterative approaches to mitigate label noise by refining Pseudo-Labels (PLs) such as [2, 3] can be intimidating and cumbersome as they require more memory and computations. Contrary to these methods and to approaches [4, 15] based on retraining from scratch on final cleansed datasets, AdaptiveDrop [16] integrates both label noise correction and filtering mechanisms in a single

framework throughout the whole training to enhance robustness against noisy labels in one stage of end-to-end training. Indeed, AdaptiveDrop adopts an adaptive and dynamic strategy to keep adjusting filtering decisions throughout training: reprocess cleansed samples to recover them in later epochs in case of wrong drop out decisions or to remove inaccurate labels that were mistakenly included in training. On the other hand, flat minima have proven recently to be effective for improving generalization and to considerably enhance performance against label noise [17, 18]. Additionally, optimizing only the training loss value, as is commonly done, can easily lead to suboptimal solutions. Indeed, learning algorithms motivated by sharpness of the loss surface, as an effective measure of generalization gap [19], have provided state-of-the-art (SOTA) performance across various tasks and were demonstrated to enhance robustness against label noise.

Motivated by these recent developments, in this paper we present the first comprehensive study of various recent optimizers for the tasks of supervised and self-supervised SV for improved generalization and robustness against label noise. We study a wide range of recent general-purpose optimizers such as ADOPT and AdEMAMix, and sharpness-based optimizers (e.g., SAM [19], Adaptive SAM [20], GAM [17], GSAM [21]) to analyze their impact on robustness of SV systems, especially when label noise is present during training.

As a comparison, we also explore different regularization methods and their combinations with optimizers for better label noise robustness, namely weight decay, exponential moving average of model parameters (EMA) [22] and SWITCH EMA [23] to learn flat optima for better generalizations without extra optimization cost. We extend our study to other common successful techniques such as label smoothing regularization [24] and mixup [25] to dilute label noise, and using small batch sizes [18] that can lead to flatter local minima to boost the performance of our networks.

Our experimental results show that the choice of the right optimizer can make a big difference in terms of training robustness to label noise and the final generalization performance. Besides, our approach adopted with our selected list of optimizers achieves superior results compared to a wide range of self-supervised SV benchmarks.

The contributions of this paper are as follows:

- We perform the first comprehensive study of a wide range of optimizers for the self-supervised SV task, as alternatives to the commonly used Adam and SGD optimizers.
- A significant number of experimental results demonstrate the effectiveness and robustness of our method under real-world label noise and various loss functions and configurations.
- Despite the effectiveness and robustness of these recent optimizers, especially under considerable label noise, we show

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that filtering label noise is still important to avoid overfitting noisy labels and to further boost generalization performance.

- Our approach outperformed numerous self-supervised SV baselines and achieved high SV performance.

2. Related Work and Motivation

Label Noise Robustness One common way to leverage large unlabeled datasets for SV systems is to use clustering models [26–28], or to employ SSL-based objectives (SimCLR, MoCo) [29] to generate PLs and train the speaker embedding network using these labels in a discriminative fashion [2, 3]. However, clustering accuracy remains a limiting factor that constrains SV performance [2, 30]. Alternatively, iterative clustering-classification methods [2, 31] aim to jointly improve PLs and SV, but errors in PLs may still propagate, reducing overall performance. Thus, there is a need for more resilient training approaches to label noise to minimize its impact on generalization.

To tackle label noise, noise-robust algorithms [32, 33] can be used to learn directly from noisy labels, or use label-cleansing approaches [14, 34] that remove or correct mislabeled samples. For instance, Subcenter-ArcFace [15] introduced a new loss function that relaxes ArcFace’s intra-class constraint, improving robustness to label noise by forming a dominant sub-class of majority clean samples and grouping hard or noisy samples into non-dominant sub-classes.

To detect mislabeled instances, one can simply employ ad-hoc anomaly measures such as cosine similarity to filter out low-quality instances or correct those that are above a certain threshold [13, 14] (e.g. low cosine similarity, exceptionally high training loss value). AdaptiveDrop employs sub-centers to dynamically track dominant sub-centers for each class from the beginning of training via a dictionary table, and performs a dynamic drop out of noisy samples at each training step from an early stage based on the latter table in order to learn progressively cleaner dominant sub-centers without the need to retrain models from scratch or to store embeddings and similarities for all samples. In the context of self-supervised speaker recognition, to mitigate the negative effect of noise in PLs, [4] proposed an audio-visual two-step label noise detection and filtering method by dividing data into easy and peculiar (hard or noisy). [2] also proposed a two-stage iterative loss-gated learning strategy where clustering of generated embeddings is applied to generate PLs, and samples with a large training loss are dropped out.

In another line of work, the BoundaryFace [35] loss adopts a label correction scheme to correct wrong labels. For that, before computing the loss of each batch, it performs label correction by deciding whether to correct a label based on whether the sample is distributed within the decision boundary of the nearest negative class: If $\max\{\cos(\theta_k + m), \forall k \neq y_i\} - \cos(\theta_{y_i}) > 0$: $y_i = k$, otherwise label y_i remains the same. y_i are labels.

Optimization Algorithms and Regularizers With today’s non-convex and often complex training loss landscapes, the training loss provides only few guarantees on model generalization ability. To improve that, Sharpness-Aware Minimization (SAM) [19] navigates the multiplicity of local and global minima by simultaneously minimizing loss value and loss sharpness. To this aim, SAM seeks parameters that lie in neighborhoods having uniformly low loss by minimizing the worst-case loss within a perturbation radius. This results in better generalization and enhanced label noise robustness. To improve sensitivity to parameter re-scaling which weakens the connection between sharpness and generalization gap, Adaptive SAM (ASAM) [20] has also introduced a concept of adaptive sharpness which is

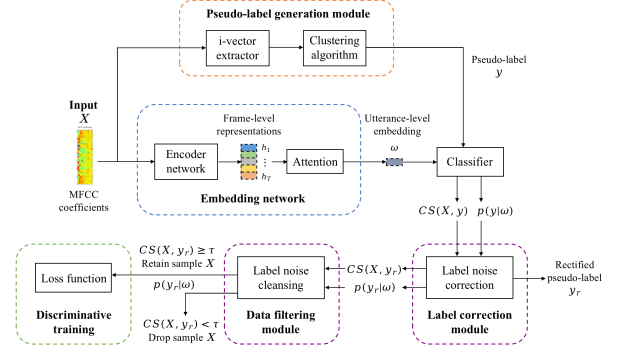


Figure 1: *General process for training our self-supervised speaker embedding network using the AdaptiveDrop framework [16]. The pipeline depicts the data flow and the label correction module employed for correcting noisy wrong labels, followed by the data filtering module used to remove the remaining misclassified samples. CS denotes the cosine similarity between the generated sample embedding ω and the class identity center embedding ω_{y_i} .*

scale-invariant. Besides, since SAM does not always prefer flat minima, GSAM [21] defines a surrogate gap to encourage more flat minima. GAM [17] also presents a novel training procedure named Gradient norm Aware Minimization to seek minima with uniformly small curvature across all directions by introducing a concept of first-order flatness as a stronger measure of flatness.

Finally, ADOPT [36] is a new adaptive gradient method, suggested as an improvement to Adam by removing the current gradient from the second moment estimate and changing the order of the momentum update and the normalization by the second moment estimate to improve convergence. Besides, to improve the contribution of older gradients in Momentum based optimizers, AdEMAMix [37] leverages two EMAs to better take advantage of past gradients and improve Adam performance.

Additionally, the mixup [25] strategy can be employed as an efficient strategy to augment data by interpolating different data samples alongside their labels to mitigate the memorization effects of label noise. In this paper, we explore two variants of mixup at both the instance input-level (i-mix) [38] and the latent space (l-mix) [27]. Moreover, Exponential Moving Average (EMA) [22] is also a widely used weight averaging regularization to learn flat optima for better generalizations without additional cost in terms of network optimization. EMA can often contribute to enhance the stability and output distribution of deep models. To take both flatness and sharpness into account during optimization, Switch EMA (SEMA) method [23] was also proposed as an enhancement to EMA by simply switching the EMA parameters to the original model after each epoch.

3. Our self-supervised SV Approach

Clustering-driven PLs can contain a considerable amount of wrongly annotated samples. To mitigate the negative effect of these wrong labels on generalization during training and to improve their accuracy via label correction, we adopt the AdaptiveDrop filtering scheme framework [16]. A detailed schematic diagram of AdaptiveDrop is presented in Figure 1 which depicts both the label filtering and correction modules.

Typically, we use Cosine Similarity (CS) as a proxy label confidence indicator to detect the mislabeled instances and set a constant CS threshold τ for label filtering to drop high-confident

noisy data. First, given each training sample x and its label y , we use the weights of our currently trained embedding network to extract embedding ω of x on the fly at each training step. We then compute cosine similarity $CS(x, y) = \cos(\omega, \omega_y) = \frac{\omega \cdot \omega_y}{\|\omega\| \|\omega_y\|}$ between ω and the current learnt class center/prototype embedding ω_y corresponding to class y . If $CS(x, y) \geq \tau$, we retain sample x in the training batch, otherwise it is dropped out from the batch. We also perform dropping from a very early stage of training to mitigate overfitting. Therefore, this can induce converging to a better trade-off between the optimized training accuracy of noisy labels and the CS to class centers.

Here in our current experiments, we adopt the angular additive margin softmax (ArcFace) loss [39] to train our models and employ the variety of aforementioned optimizers to learn weights and compare their performances.

The ArcFace objective is formulated as follows:

$$L_{ArcFace} = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{e^{s(\cos(\theta_{y_i} + m))}}{K_1} \right), \quad \text{where}$$

$$K_1 = e^{s(\cos(\theta_{y_i} + m))} + \sum_{j=1, j \neq i}^C e^{s \cos \theta_j},$$

N is the batch size, C is the number of classes, y_i corresponds to label index, θ_{y_i} represents the angle between the column vector of weight matrix $W_{y_i} \in \mathbb{R}^{512 \times 1}$ of the class center and the i^{th} feature embedding $x_i \in \mathbb{R}^{512 \times 1}$, where both W_{y_i} and x_i are l_2 normalized. $\theta_j = \arccos(W_j^T x_i)$ is the angle between x_i and the j^{th} class center $W_j \in \mathbb{R}^{512 \times 1}$. The scale factor s prevents the gradient from becoming too small during training, and m is a margin to ensure correct classes have higher similarity than incorrect ones.

For label correction, we adopt the label correction scheme of BoundaryFace [35] (without its proposed regularization term to emphasize hard samples). Additionally, inspired from the idea of sub-centers [15] used in [5], we employ a max pooling step on the subclass-wise cosine similarity scores ($W_j^T x_i$) across all our predefined K sub-centers. This helps to identify dominant clean sub-centers to use later as class prototypes for enhanced label noise robustness. Given a weight matrix $W \in \mathbb{R}^{C \times K \times 512}$ of all sub-centers, the Subcenter-ArcFace objective that we now use is formulated as follows: $L_{Subcenter-ArcFace} = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{e^{s(\cos(\theta_{i,j} + m))}}{K_1} \right)$, where now $K_1 = e^{s(\cos(\theta_{i,j} + m))} + \sum_{j=1, j \neq i}^C e^{s \cos \theta_{i,j}}$, and $\theta_{i,j} = \arccos(\max_k(W_{j,k}^T x_i))$ for $k \in 1, \dots, K$.

To this purpose, we employ a sub-centers dictionary table DSCT $\in \mathbb{N}^{C \times K}$ (initially all entries are set to zero) to adaptively track the dominant ones per class from the beginning of training. As a rule, the ongoing sub-centers with the highest value per class are considered dominant (see Algorithm 1). Algorithm 1 shows our implementation of AdaptiveDrop using sub-centers.

4. Experimental setup

We conducted a set of experiments based on the VoxCeleb2 dataset [7]. To train the embedding networks, we used the development subset of the VoxCeleb2 dataset, which consists of 1,092,009 utterances collected from 5,994 speakers. The evaluation was performed according to the Original (Vox1-O) Vox-Celeb1 trials lists [6].

We employ ECAPA-TDNN [40] as our speaker embedding network. As acoustic features for our SV experiments, we used 40-dimensional Mel-frequency cepstral coefficients (MFCCs) extracted at every 10 ms, using a 25 ms Hamming window via Kaldi toolkit [41]. Moreover, we have used waveform-level data augmentations including additive noise and room impulse response (RIR) simulation [42] to follow other SV works. Besides, we have applied augmentation over the extracted MFCCs

Algorithm 1 AdaptiveDrop Algorithm (using Sub-centers)

Inputs: Training data X , noisy pseudo-labels Y , model M , loss function l , cosine similarity threshold τ , epoch to start correcting noisy labels (ESC), epoch to start dropping out noisy labels (ESD), epoch to start tracking dominant sub-centers (ESTD), dominant sub-centers table DSCT $\in \mathbb{N}^{C \times K}$, optimizer Opt.

Output: The best model M .

Initialize: Table DSCT with zeros

for epoch = 1 **to** max_epochs **do**

 Extract embeddings ω for samples of current batch x .

 Compute cosine similarities (CS) between each sample in x and its corresponding class sub-centers $\omega_{y,j}$ for $j \in \{1, \dots, K\}$.

if epoch $\geq$ ESTD **then**

 #Update dominant sub-centers table (DSCT)

 closest_subcenter = $\arg \max_{j \in \{1, \dots, K\}} CS[y, j]$

 DSCT [y , closest_subcenter] += 1

end if

if epoch $\geq$ ESC **then**

 Correct pseudo-labels y of batch x .

end if

if epoch $\geq$ ESD **then**

 #Compute dominant sub-centers

$K_{\text{dominant}} = \arg \max_{i \in \{1, \dots, K\}} \text{DSCT}[y, i]$

 Filter samples based on $CS(x, y) = \cos(\omega, \omega_{y, K_{\text{dominant}}})$.

 Set $(x_{\text{clean}}, y_{\text{clean}}) = \{(x_i, y_i) \text{ for } x_i, y_i \in (x, y) \text{ if } CS > \tau\}$

else

 Set $x_{\text{clean}} = x$ and $y_{\text{clean}} = y$

end if

 Compute loss l with filtered x_{clean} and y_{clean} .

 Perform gradient descent to update parameters of model M using Opt.

end for

features, analogous to the specaugment scheme [43]. All SV experiments have been run for 150 epochs using a single A40 GPU (around 2 days training), with a default batch size of 200 MFCC samples. Scale factor $s = 30$ and margin $m = 0.2$ were used across all margin-based losses. Besides, we do not use score normalization and CS was used as a backend for verification scoring between enrollment and test embeddings. We refer to our loss implementation using label correction with BoundaryFace and refer to sub-centers combined with label correction with subcenter-BoundaryFace.

Finally, we use a default $K = 3$ sub-centers when applied and a CS threshold of $\tau = 0.423$. For AdaptiveDrop, use ESD = 5, ESC = 7, ESTD = 3. Besides, to avoid training instability especially for highly noisy thresholds, we only drop out a maximum of 50% of samples from each training batch. Additionally, we employ a first very short warm-up stage of training for the first 5 epochs to train our embedding network with all the PLs to have meaningful weights when label noise filtering and correction begin. Besides, we use the CAMSAT clustering algorithm [44] to generate PLs using 10000 as a predefined number of clusters. We use $\lambda = 0.001$ for weight decay regularization, and a default smoothing coefficient ϵ of 0.15. Code is available at <https://github.com/fathana/Optimizers>

5. Results and Discussion

First of all, to assess the effectiveness of our studied optimizers and regularization techniques, in Tables 1 and 2 we compare the performance of our system under different configurations, trained with different optimizers. In Table 1, we report the SV performance using ground truth labels, while in Table 2 we use CAMSAT-based PLs (70.9% unsupervised clustering accuracy) to train our systems. Results show that sharpness and flatness-based optimizers can be effective to improve generalization and robustness of SV systems under label noise. We find these improvements more pronounced when labels are noisy than when training with true labels. We also notice that these SAM-based optimizers are considerably more effective than most of our other studied regularization techniques.

Table 1: Supervised SV performance in terms of EER (%) on the Vox1-O test set, using ground-truth labels across various optimizers and configurations. When unspecified, Adam is used.

Optimizer / Regularization	ArcFace	Sub-center ArcFace
Adam	1.437	1.384
SGD	2.63	2.572
SAM+SGD	3.728	2.921
ASAM+SGD	2.651	3.112
SAM+Adam	1.448	1.373
ASAM+Adam	1.41	1.363
GAM+Adam	1.331	1.204
GSAM+Adam	1.22	1.214
ADOPT	1.341	1.41
AdEMAMix	1.474	-
AdamW (weight decay)	1.538	1.543
EMA	1.379	1.283
SEMA	1.389	1.379
i-mix	1.294	1.432
l-mix	1.458	1.341
Label smoothing	1.596	1.304
i-mix + Label smoothing	-	1.384
l-mix + Label smoothing	-	1.564
BoundaryFace	1.341	1.315
i-mix + BoundaryFace	1.469	1.67
l-mix + BoundaryFace	1.638	1.543
Label smoothing + BoundaryFace	1.469	1.267

Table 2: SV perf. (EER) on the Vox1-O test set, using CAMSAT-based PLs across various optimizers and configurations.

Optimizer	Regularization techniques	Batch size	EER (%)
Adam	-	50	2.407
Adam	-	200	2.508
ASAM+Adam	-	50	1.983
SAM+Adam	-	200	2.439
Adam	weight decay	50	2.63
GSAM+Adam	-	50	2.057
ASAM+Adam	SubcenterBoundaryFace	200	2.365
ASAM+Adam	BoundaryFace, no filtering	200	2.603
ASAM+Adam	BoundaryFace	200	2.375
Adam	SubcenterBoundaryFace	50	2.625
Adam	l-mix, no filtering	200	3.033
Adam	i-mix, no filtering	200	3.139
ASAM+Adam	Label smoothing ($\epsilon = 0.2$)	50	2.073

Table 3: SV performance in terms of EER (%) on the Vox1-O test set, using CAMSAT-based PLs with and without Adam -based ASAM optimizer across various configurations.

Loss function	Regularization techniques	EER (%)	
		without ASAM+Adam	with ASAM+Adam
ArcFace (supervised)	-	1.463	1.41
Subcenter-ArcFace (supervised)	-	1.384	1.363
ArcFace	-	3.134	3.001
Subcenter-ArcFace	-	2.943	2.656
Subcenter-ArcFace	i-mix	3.139	2.725
Subcenter-ArcFace	l-mix	3.033	2.89
Subcenter-ArcFace	AdaptiveDrop	2.407	1.983
BoundaryFace	-	2.752	2.503
BoundaryFace	AdaptiveDrop	2.513	2.413
Subcenter-BoundaryFace	AdaptiveDrop	2.672	2.338

Table 4: SV performance in terms of EER (%) on the Vox1-O test set, using CAMSAT-based PLs with and without AdaptiveDrop noise filtering across various optimizers (batch size is 200).

Loss function	Optimizer	EER (%)	
		without AdaptiveDrop	with AdaptiveDrop
BoundaryFace	ASAM+Adam	2.556	2.37
Subcenter-ArcFace	ASAM+Adam	2.656	2.153
Subcenter-ArcFace	SAM+Adam	2.906	2.439
Subcenter-ArcFace	SGD	3.256	3.006
Subcenter-ArcFace	ASAM	2.996	2.789
Subcenter-ArcFace	SAM	3.192	2.704
Subcenter-ArcFace	AdEMAMix	2.895	2.715
Subcenter-ArcFace	Adam	2.943	2.407
Subcenter-ArcFace	AdamW (weight decay)	2.869	2.678
Subcenter-ArcFace	Adopt	41.803	3.728

Table 5: Some recent SOTA self-supervised SV approaches in EER (%) compared to AdaptiveDrop on the Vox1-O eval set.

SSL Objective	EER (%)
MoBY [29]	8.2
InfoNCE [2]	7.36
MoCo [45]	7.3
ProtoNCE [29]	7.21
PCL [29]	7.11
CA-DINO [46]	3.585
i-mix [47]	3.478
l-mix [47]	3.377
Iterative clustering [2]	3.09
CAMSAT [44]	3.065
Subcenter-BoundaryFace [48]	2.752
AdaptiveDrop [16]	2.407
Proposed (ours) [with AdaptiveDrop & ASAM+Adam]	1.983

Additionally, in Table 3 we perform further ablations using different configurations to assess the effectiveness of the Adam-based ASAM optimizer (denoted as ASAM+Adam). Besides, in Table 4 we assess the contribution of AdaptiveDrop label noise filtering compared to our best performing ASAM+Adam sharpness-based optimizer. Results show that ASAM+Adam optimizer is effective across all systems. However, Table 4 shows that label noise filtering is even more effective than sharpness optimization, and that label filtering and loss sharpness minimization of SAM are complementary to each other.

Finally, Table 5 shows a comparison of our approach using our best-performing configuration with Subcenter-ArcFace, AdaptiveDrop filtering, Adam-based ASAM optimizer, and CAMSAT-based PLs, compared to recent SOTA self-supervised SV approaches employing diverse SSL objectives with the same ECAPA-TDNN model encoder, on Vox1-O. The results show clearly that our approach provides large performance gains compared to all baselines, while being simple, fast and performed end-to-end in a single stage of training.

6. Conclusion

In this work, we investigate the performance of various recent optimizers and regularization methods to improve robustness of self-supervised (SS) training in the presence of noisy labels. We proposed a framework encompassing label noise filtering, label correction, Sharpness-Aware Minimization, and sub-centers of classes to integrate with any type of loss objective. Besides, our extensive experiments show that the choice of optimizer can be crucial for final performance, that label filtering is paramount, and that our method outperforms various other SS speaker verification benchmarks.

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8. References

- [1] J. H. Hansen *et al.*, “Speaker recognition by machines and humans: A tutorial review,” *IEEE Signal Processing Magazine*, 2015.
- [2] R. Tao *et al.*, “Self-supervised speaker recognition with loss-gated learning,” in *ICASSP*. IEEE, 2022.
- [3] J. Peng *et al.*, “Progressive Contrastive Learning for Self-Supervised Text-Independent Speaker Verification,” in *Proc. of Odyssey Workshop*, 2022.
- [4] R. Tao *et al.*, “Speaker recognition with two-step multi-modal deep cleansing,” in *IEEE ICASSP 2023*. IEEE, 2023, pp. 1–5.
- [5] A. Fathan, X. Zhu, and J. Alam, “On the influence of regularization techniques on label noise robustness: Self-supervised speaker verification as a use case,” in *Proc. of the IEEE IJCB, Buffalo, New York, USA, 15-18 September 2024*.
- [6] A. Nagrani, J. S. Chung *et al.*, “Voxceleb: a large-scale speaker identification dataset,” in *INTERSPEECH*, 2017.
- [7] J. S. Chung, A. Nagrani, and A. Zisserman, “Voxceleb2: Deep speaker recognition,” in *INTERSPEECH*, 2018.
- [8] D. Arpit, S. Jastrzebski, N. Ballas, D. Krueger *et al.*, “A closer look at memorization in deep networks,” in *International conference on machine learning*. PMLR, 2017, pp. 233–242.
- [9] M. N. Rizve *et al.*, “In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning,” *arXiv preprint arXiv:2101.06329*, 2021.
- [10] H. Wei *et al.*, “Mitigating memorization of noisy labels by clipping the model prediction,” in *International Conference on Machine Learning*. PMLR, 2023, pp. 36 868–36 886.
- [11] B. Frénay, A. Kabán *et al.*, “A comprehensive introduction to label noise,” in *ESANN*. Citeseer, 2014.
- [12] I. Guyon, N. Matic, V. Vapnik *et al.*, “Discovering informative patterns and data cleaning,” 1996.
- [13] J.-w. Sun *et al.*, “Identifying and correcting mislabeled training instances,” in *Future generation communication and networking (FGCN 2007)*, vol. 1. IEEE, 2007, pp. 244–250.
- [14] C. E. Brodley and M. A. Friedl, “Identifying mislabeled training data,” *Journal of artificial intelligence research*, 1999.
- [15] J. Deng, J. Guo *et al.*, “Sub-center arcface: Boosting face recognition by large-scale noisy web faces,” in *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, Proceedings, Part XI 16*. Springer, 2020, pp. 741–757.
- [16] A. Fathan, X. Zhu, and J. Alam, “Adaptivedrop: A simple adaptive label noise filtering scheme for enhanced self-supervised speaker verification,” in *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*.
- [17] X. Zhang *et al.*, “Gradient norm aware minimization seeks first-order flatness and improves generalization,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 20 247–20 257.
- [18] N. S. Keskar *et al.*, “On large-batch training for deep learning: Generalization gap and sharp minima,” *arXiv preprint arXiv:1609.04836*, 2016.
- [19] P. Foret *et al.*, “Sharpness-aware minimization for efficiently improving generalization,” *arXiv preprint arXiv:2010.01412*, 2020.
- [20] J. Kwon *et al.*, “Asam: Adaptive sharpness-aware minimization for scale-invariant learning of deep neural networks,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 5905–5914.
- [21] J. Zhuang *et al.*, “Surrogate gap minimization improves sharpness-aware training,” *arXiv preprint arXiv:2203.08065*, 2022.
- [22] B. T. Polyak and A. B. Juditsky, “Acceleration of stochastic approximation by averaging,” *SIAM journal on control and optimization*, vol. 30, no. 4, pp. 838–855, 1992.
- [23] S. Li *et al.*, “Switch ema: A free lunch for better flatness and sharpness,” *arXiv preprint arXiv:2402.09240*, 2024.
- [24] C. Szegedy *et al.*, “Rethinking the inception architecture for computer vision,” in *Proc. of the IEEE conference on CVPR*, 2016, pp. 2818–2826.
- [25] H. Zhang *et al.*, “mixup: Beyond empirical risk minimization,” *arXiv preprint arXiv:1710.09412*, 2017.
- [26] A. Fathan, J. Alam, and W. Kang, “On the impact of the quality of pseudo-labels on the self-supervised speaker verification task,” in *NeurIPS 2022 Second ENLSP Workshop*, 2022. [Online]. Available: https://neurips2022-enlsp.github.io/papers/paper_51.pdf
- [27] W. H. Kang *et al.*, “L-mix: A latent-level instance mixup regularization for robust self-supervised speaker representation learning,” *IEEE JSTSP*, vol. 16, no. 6, pp. 1263–1272, 2022.
- [28] W. H. Kang, J. Alam, and A. Fathan, “An analytic study on clustering-based pseudo-labels for self-supervised deep speaker verification,” in *SPECOM*, 2022.
- [29] W. Xia *et al.*, “Self-supervised text-independent speaker verification using prototypical momentum contrastive learning,” in *ICASSP*. IEEE, 2021.
- [30] B. Han, Z. Chen, and Y. Qian, “Self-supervised speaker verification using dynamic loss-gate and label correction,” *arXiv preprint arXiv:2208.01928*, 2022.
- [31] Y. Li *et al.*, “Contrastive clustering,” in *AAAI*, 2021.
- [32] M. Guan *et al.*, “Who said what: Modeling individual labelers improves classification,” in *Proc. of AAAI*, vol. 32, no. 1, 2018.
- [33] D. Rolnick, A. Veit *et al.*, “Deep learning is robust to massive label noise,” *ICLR*, 2018.
- [34] D. T. Nguyen *et al.*, “Self: Learning to filter noisy labels with self-ensembling,” *arXiv preprint arXiv:1910.01842*, 2019.
- [35] S. Wu, “Boundaryface: A mining framework with noise label self-correction for face recognition,” in *ECCV*. Springer, 2022.
- [36] S. Taniguchi *et al.*, “Adopt: Modified adam can converge with any β_2 with the optimal rate,” *arXiv preprint arXiv:2411.02853*, 2024.
- [37] M. Pagliardini *et al.*, “The ademanix optimizer: Better, faster, older,” *arXiv preprint arXiv:2409.03137*, 2024.
- [38] K. Lee *et al.*, “i-mix: A domain-agnostic strategy for contrastive representation learning,” in *ICLR*, 2021.
- [39] J. Deng *et al.*, “Arcface: Additive angular margin loss for deep face recognition,” *IEEE TPAMI*, 2021.
- [40] B. Desplanques *et al.*, “ECAPA-TDNN: emphasized channel attention, propagation and aggregation in TDNN based speaker verification,” in *Interspeech 2020*. ISCA.
- [41] D. Povey *et al.*, “The kaldi speech recognition toolkit,” in *In IEEE 2011 workshop*, 2011.
- [42] D. Snyder *et al.*, “X-vectors: Robust dnn embeddings for speaker recognition,” in *Proc. of IEEE ICASSP*, 2018, pp. 5329–5333.
- [43] D. S. Park *et al.*, “Specaugment: A simple data augmentation method for automatic speech recognition,” in *Interspeech*, 2019, pp. 2613–2617.
- [44] A. Fathan and J. Alam, “Camsat: Augmentation mix and self-augmented training clustering for self-supervised speaker recognition,” in *IEEE ASRU Workshop*, 2023.
- [45] J. Cho *et al.*, “The jhu submission to voxsrc-21: Track 3,” *arXiv preprint arXiv:2109.13425*, 2021.
- [46] B. Han *et al.*, “Self-supervised learning with cluster-aware-dino for high-performance robust speaker verification,” *arXiv preprint arXiv:2304.05754*, 2023.
- [47] A. Fathan and J. Alam, “On the influence of the quality of pseudo-labels on the self-supervised speaker verification task: a thorough analysis,” in *IWBF*. IEEE, 2023.
- [48] A. Fathan, X. Zhu, and J. Alam, “An investigative study of the effect of several regularization techniques on label noise robustness of self-supervised speaker verification systems,” *Odyssey 2024*.